Practice workbook · Wired or wireless? Media, pros and cons, and network speed units

Advantages and drawbacks of wired versus wireless links, the transmission media used, Wi-Fi frequency bands, units of network speed (bits per second), and choosing a private hotspot over public Wi-Fi.

7 question parts: 1 worked-example retry and 6 further practice parts.
Name: _____ Class: _____ Date: _____

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

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Type ID: CSC-T56-02

1. Question 4 · part c.i · 2 marks

2023 JUN · Paper 01

Worked-example retry · Attempt part c.i only.

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4 Data packets travel across networks from one device to another.

(a) Identify the measurement of network speeds.

(1)

- ☒ **A** Mebibits per second
- ☒ **B** Mebibytes per second
- ☒ **C** Megabits per second
- ☒ **D** Megabytes per second

(b) Data packets contain the addresses of the sender and the receiver.

Complete the table to give the number of bits that make up each type of network address.

(2)

Type	Example	Number of bits
IPv4	192.169.0.3	
IPv6	1050:a500:00c0:0440:0006:0300:700d:436f	

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(c) Data packets travel over physical media.

(i) Describe **one** difference between the media used by a wired network and a wireless network.

(2)

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.....

.....

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1. Question 4 · part c.i · 2 marks

2023 JUN · Paper 01

Worked-example retry · Attempt part c.i only.

(ii) Some people confuse Ethernet® and Wi-Fi.

Describe what is meant by the term **Ethernet®**.

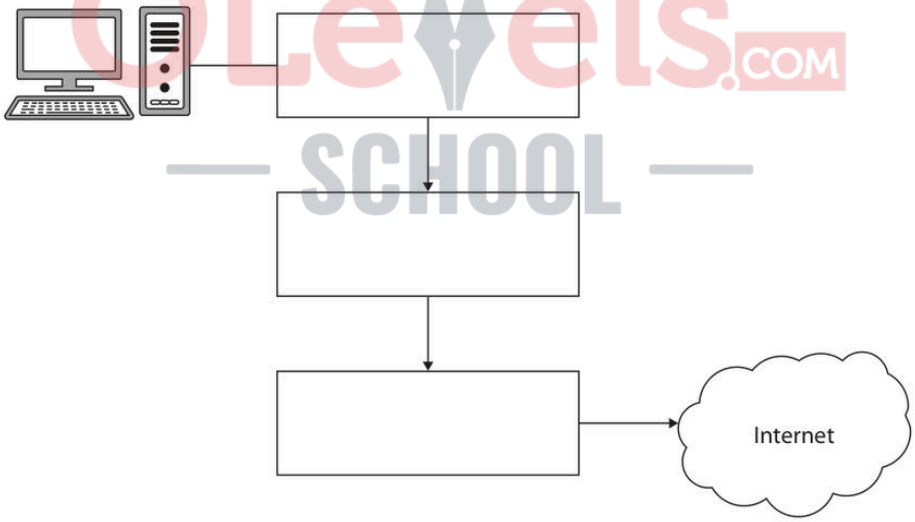
(2)

(d) A single physical box connects a desktop computer to the Internet.

The box incorporates three different components.

Complete the diagram to show the names of the components in the correct order.

(3)



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P 7 2 9 3 7 A 0 1 2 2 4

1. Question 4 · part c.i · 2 marks · Working and answer

Working and answer

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2. Question 3 · part b · 1 mark

2019 JUN · Paper 01

Further practice · Attempt part b only.

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3 Isra works in an office building where she has access to different types of network.

(a) Isra uses different types of networks for different tasks.

(i) Name the type of network she uses to print a document on the printer in the office next door.

(1)

.....

.....

(ii) Name the type of network she uses to order stationery from an online supplier.

(1)

.....

.....

(b) Identify the measurement used for network data speeds.

(1)

- ☒ **A** Megabits per second
- ☒ **B** Megapixels per second
- ☒ **C** Mebicycles per second
- ☒ **D** Mebihertz per second

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2. Question 3 · part b · 1 mark

2019 JUN · Paper 01

Further practice · Attempt part b only.

(c) Isra uses her tablet computer and smartphone to access email.

She wants to set up a new email account.

State the email protocol she should use.

Justify your choice.

(3)

Email protocol

Justification

(d) The table describes characteristics of different network topologies.

Complete the table to match each characteristic to **one** network topology.

(4)

Characteristic	Bus	Ring	Star	Mesh
Network performance degrades as more devices are added.	☒	☒	☒	☒
All devices are directly connected to all others.	☒	☒	☒	☒
Each node has a physical attachment to a routing device.	☒	☒	☒	☒
There are no data collisions, because packets travel in the same direction.	☒	☒	☒	☒

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2. Question 3 · part b · 1 mark · Working and answer

Working and answer

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3. Question 1 · part b.i · 2 marks

2020 NOV · Paper 01

Further practice · Attempt part b.i only.

1 Upeka manages the network in her office building.

(a) Upeka uses the network to install software updates on all the office computers.

Give **two other** reasons why computers are connected in a network.

(2)

1 .....

.....

2 .....

.....

(b) The network uses wired and wireless connectivity.

(i) State **two** advantages of using wired rather than wireless connectivity.

(2)

1 .....

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2 .....

.....

(ii) State **two** disadvantages of using wired rather than wireless connectivity.

(2)

1 .....

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2 .....

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3. Question 1 · part b.i · 2 marks · Working and answer

Working and answer

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4. Question 1 · part b.ii · 2 marks

2020 NOV · Paper 01

Further practice · Attempt part b.ii only.

1 Upeka manages the network in her office building.

(a) Upeka uses the network to install software updates on all the office computers.

Give **two other** reasons why computers are connected in a network.

(2)

1 .....

.....

2 .....

.....

(b) The network uses wired and wireless connectivity.

(i) State **two** advantages of using wired rather than wireless connectivity.

(2)

1 .....

.....

2 .....

.....

(ii) State **two** disadvantages of using wired rather than wireless connectivity.

(2)

1 .....

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2 .....

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4. Question 1 · part b.ii · 2 marks

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Further practice · Attempt part b.ii only.

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(c) Some people confuse the terms 'internet' and 'world wide web'.

Describe the difference between the internet and the world wide web.

(2)

(d) Networks use both IPv4 and IPv6 addresses to identify connected devices.

Explain why IPv6 addressing was introduced.

(2)

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4. Question 1 · part b.ii · 2 marks · Working and answer

Working and answer

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5. Question 3 · part c · 1 mark

2021 NOV · Paper 01

Further practice · Attempt part c only.

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3 Carlo uses his smartphone to make hands-free calls from the car.

(a) The smartphone connects to the car wirelessly forming a network.

Complete the table by adding a tick (✓) to indicate the network type and a tick (✓) to indicate the usage model the network uses.

(2)

Network type	Tick (✓)	Usage model	Tick (✓)
Local Area Network (LAN)		Client-Server	
Wide Area Network (WAN)		Peer-to-peer	
Personal Area Network (PAN)			

(b) Data on a network is transmitted between devices using packets and the TCP/IP layered protocol stack.

Complete the table by naming **two** layers of the stack and giving **one** function for each of the layers that you have named.

(4)

	Layer name	Function
1		
2		

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5. Question 3 · part c · 1 mark

2021 NOV · Paper 01

Further practice · Attempt part c only.

(c) The smartphone can communicate using Wi-Fi.

Identify which **one** of these is a radio frequency used by smartphones to connect to Wi-Fi.

(1)

- ☒ **A** 2.4 GHz
- ☒ **B** 3 KHz
- ☒ **C** 4.1 GHz
- ☒ **D** 5 KHz

(d) Carlo's smartphone can use 3G, 4G and 5G mobile communication standards.

(i) Give **one** advantage of using a higher frequency band for mobile communications.

(1)

(ii) Explain **one** benefit to Carlo of using 5G rather than 3G.

(2)

(e) Carlo wants to reduce the environmental impact of his smartphone.

(i) Give **one** way Carlo could reduce energy consumption.

(1)

(ii) Explain **one** way, other than reducing energy consumption, that Carlo could reduce the environmental impact of his smartphone.

(2)

(Total for Question 3 = 13 marks)

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5. Question 3 · part c · 1 mark · Working and answer

Working and answer

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6. Question 6 · part b.ii · 2 marks

2022 JUN · Paper 01

Further practice · Attempt part b.ii only.

6 Santiago manages a computer network for a small business.

(a) Networks are based on a topology.

Figure 3 shows a network topology.

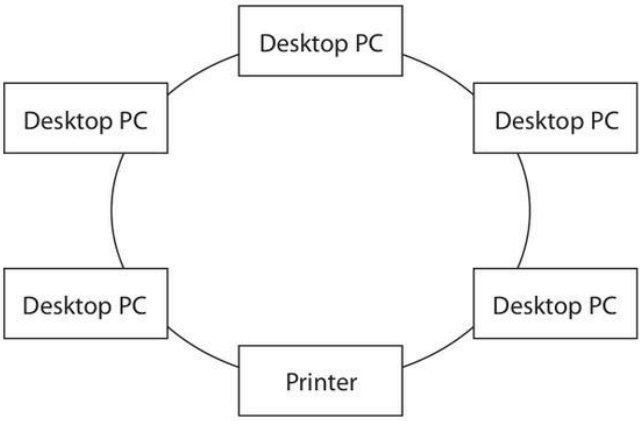


Figure 3

(i) Explain **one** benefit of this network topology. **(2)**

(ii) The internet is the world's largest mesh network.

Explain **one** reason why a mesh topology is essential for the internet. **(2)**

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6. Question 6 · part b.ii · 2 marks

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Further practice · Attempt part b.ii only.

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(b) Santiago works on his laptop whilst travelling by train.

There is a free Wi-Fi connection on the train, but Santiago doesn't use it.

He prefers to set up a network between his smartphone and his laptop to connect to the internet.

(i) Name this type of network.

(1)

(ii) Explain **one** advantage for Santiago of using the network he has set up to connect to the internet, rather than the free Wi-Fi connection.

(2)

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(c) Santiago uses audit trails to help protect the network.

(i) State what is meant by an **audit trail**.

(1)

(ii) Give **one** way the data from audit trails can be used to help keep the network secure.

(1)

6. Question 6 · part b.ii · 2 marks · Working and answer

Working and answer

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7. Question 4 · part a · 1 mark

2023 JUN · Paper 01

Further practice · Attempt part a only.

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7. Question 4 · part a · 1 mark · Working and answer

Working and answer

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